

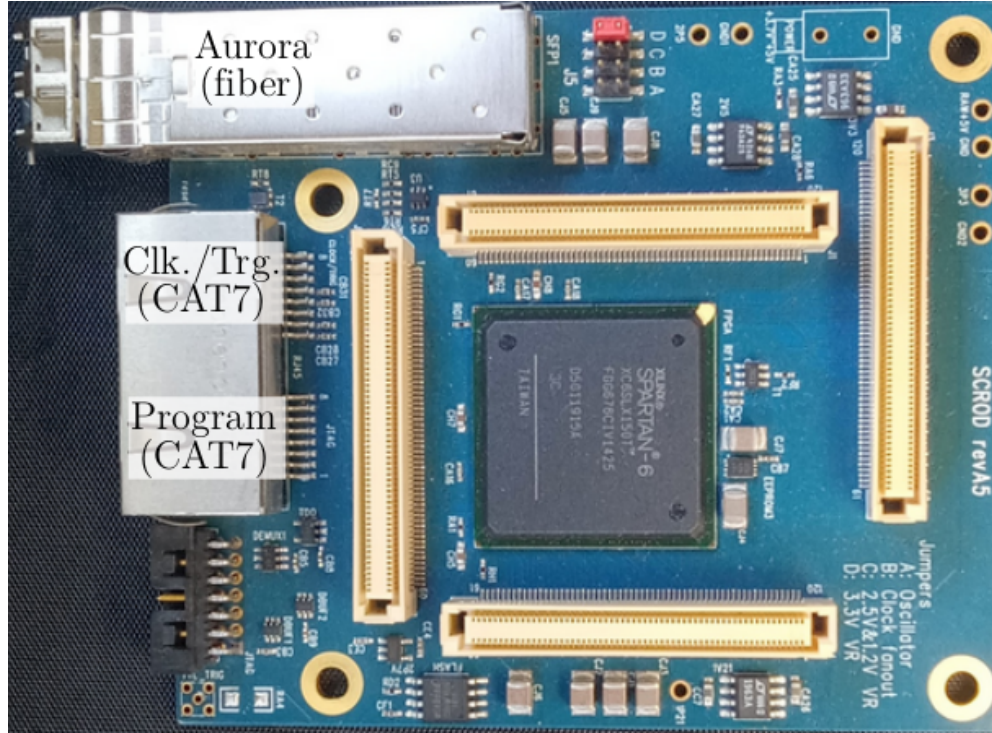
KLM Scintillator SRAM Expansion



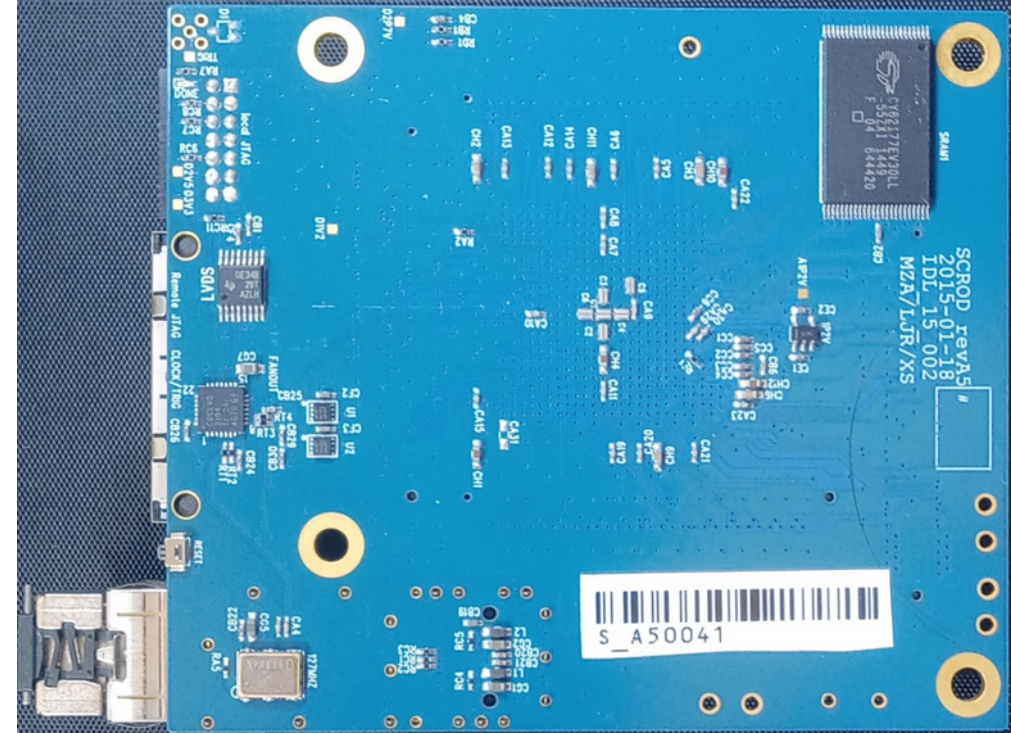
Chris Ketter & Horacio Ledesma
University of Hawaii at Mānoa



SCROD



Front

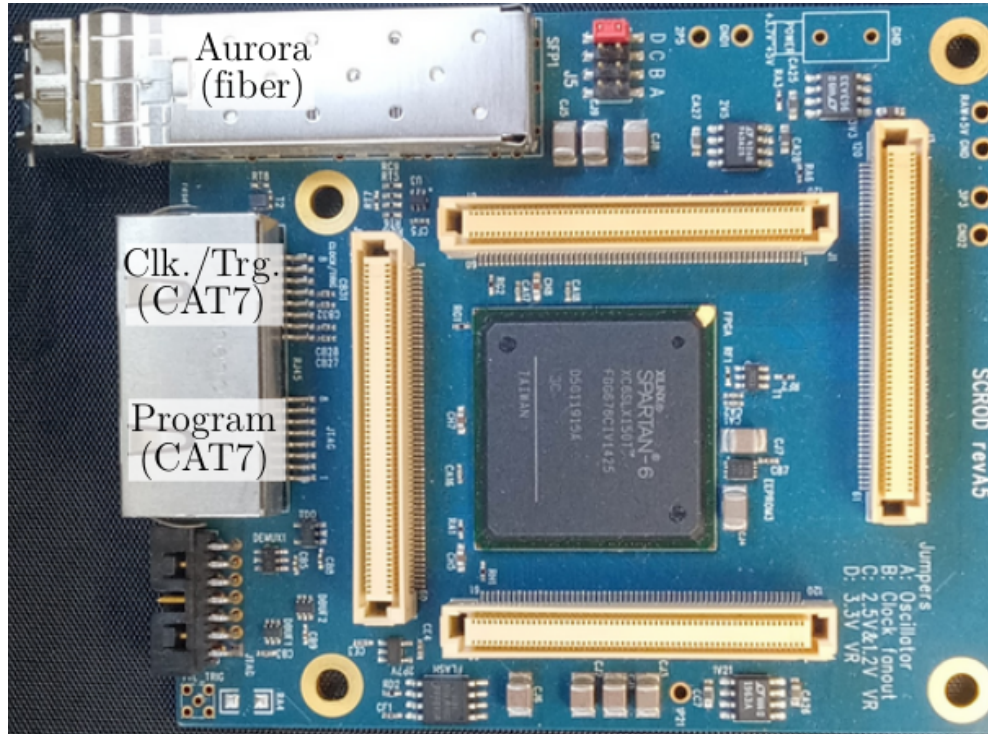


Back

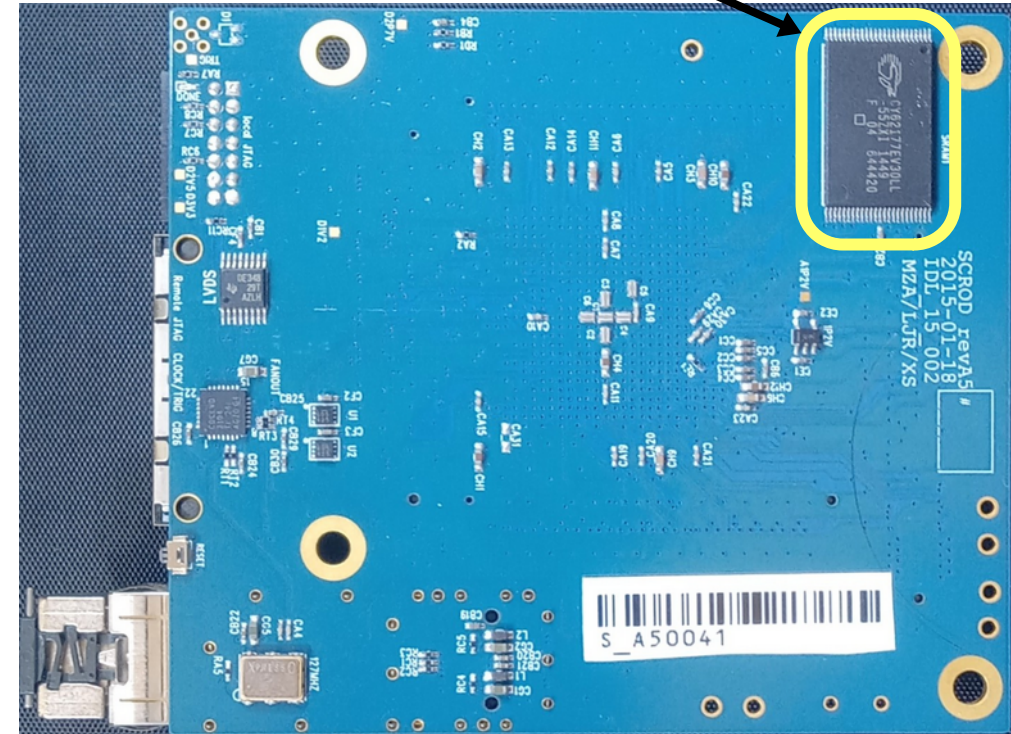
SCROD

SRAM --> 4M addr. X 8 bits wide

(one mem. ctrl. serves two --- bus A/B --- firmware modules)

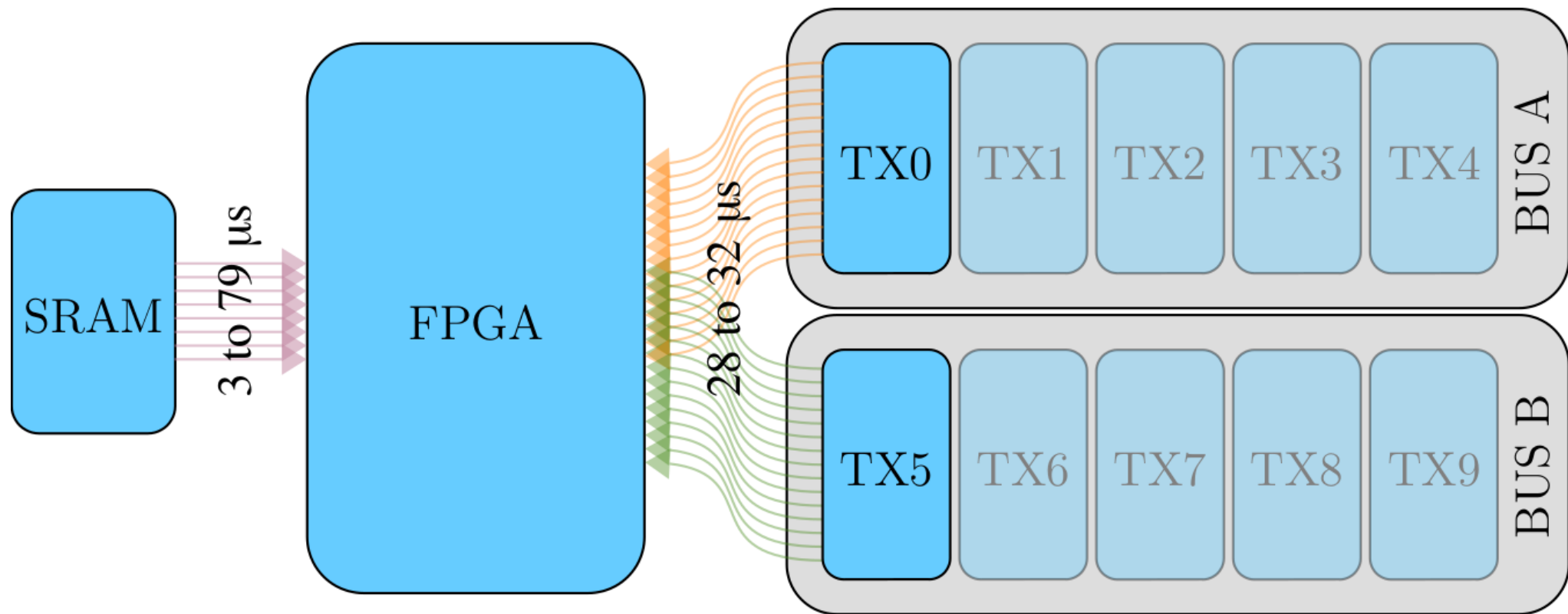


Front

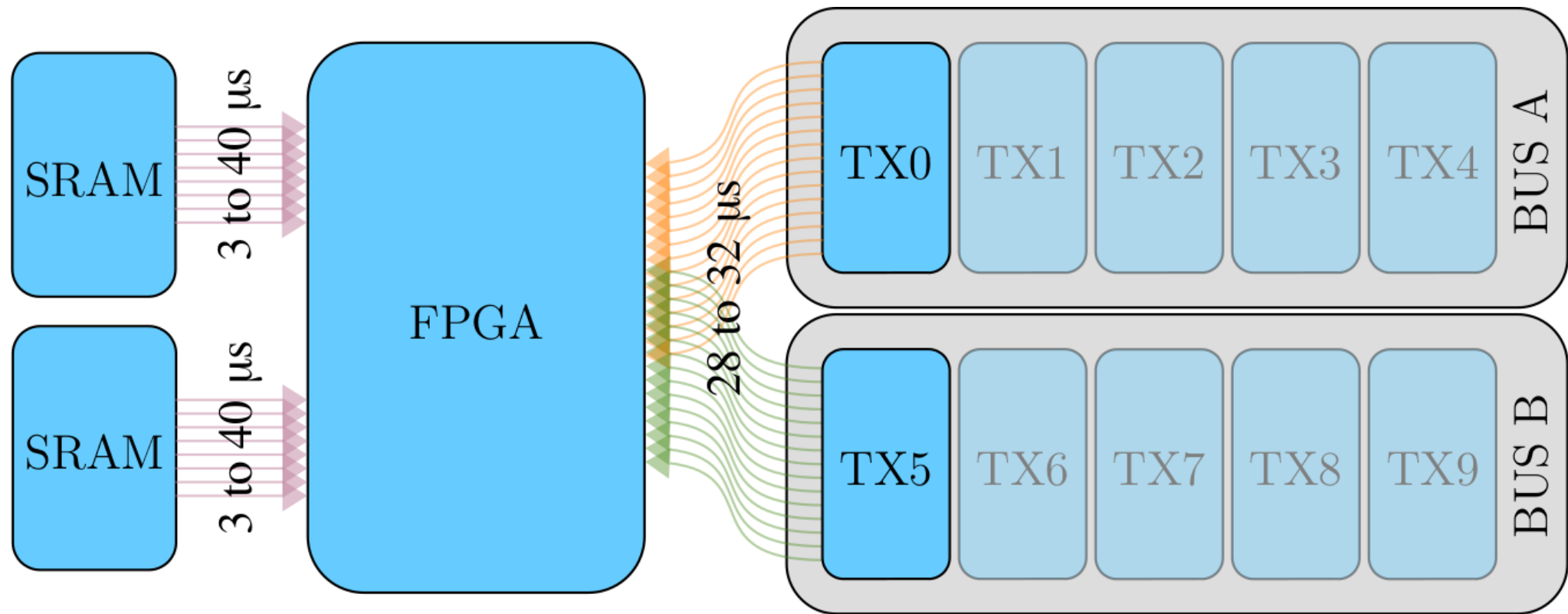


Back

BOTTLENECK



MITIGATION



SCROD SRAM EXPANSION BOARD

- ☕ Important for multi-strip hits wherein pedestal fetching becomes the bottleneck
- ☕ The US-Belle II operations budget confirmed \$15k for the project



DESIGN REQUIREMENTS

- ☕ At least 2M X 8 in memory depth
- ☕ As fast as existing SRAM
- ☕ Installation must be possible without uncabling (and thereby possibly damaging) readout electronics
- ☕ External 3.3 V power supply must meet same installation criteria

DESIGN REQUIREMENTS

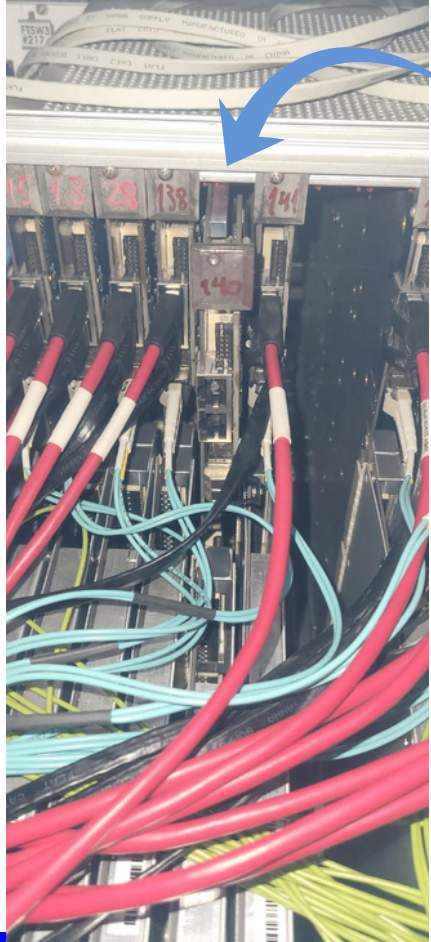
- ☕ At least 2M X 8 in memory depth
- ☕ As fast as existing SRAM
- ☕ Installation must be possible without uncabling (and thereby possibly damaging) readout electronics
- ☕ External 3.3 V power supply must meet same installation criteria

(Extra --- wish list --- item: Can we install standoffs under SCROD at the same time?

→ Recall persistent issue with SCRODs coming unseated!)

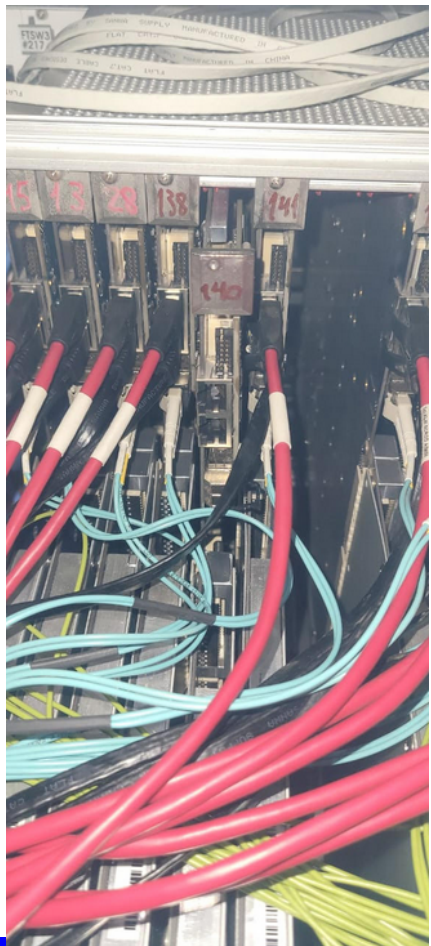
Ease of installation?

ACCESS TEST (LAST JULY)



☕ Slid out a central board in EB0

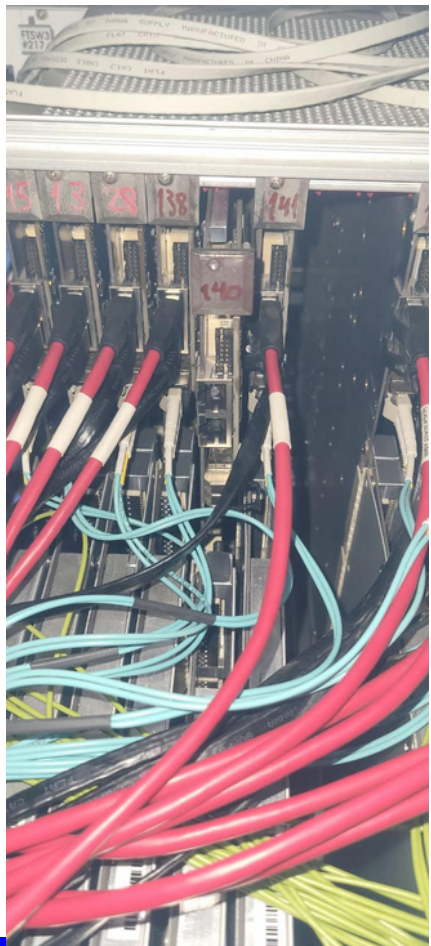
ACCESS TEST (LAST JULY)



- ☕ Slid out a central board in EB0
- ... until safety tab contacted crossbar

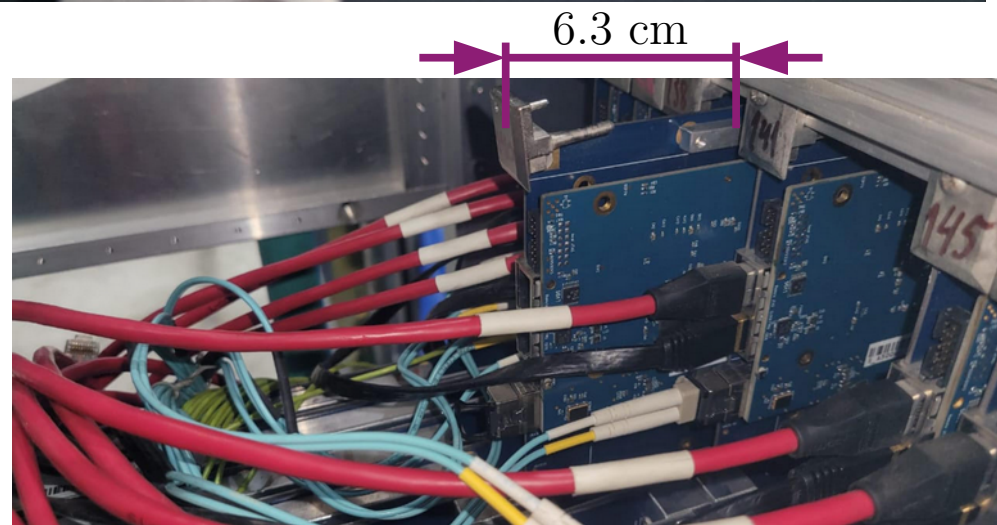


ACCESS TEST (LAST JULY)



☕ Slid out a central board in EB0
... until safety tab contacted crossbar

☕ Yields 6.3 cm of space for single-SCROD access



ACCESS TEST (LAST JULY)



- ☕ Definitely enough room to access mezzanine connector for SRAM expansion board
- ☕ Can also install **at least one** standoff between SCROD and motherboard --- two diagonally would be much much better
- ☕ **But, we had an issue with RHIC unplugging from motherboard**

TABLETOP INVESTIGATION

☕ Checked situation using broken FEEs in storage

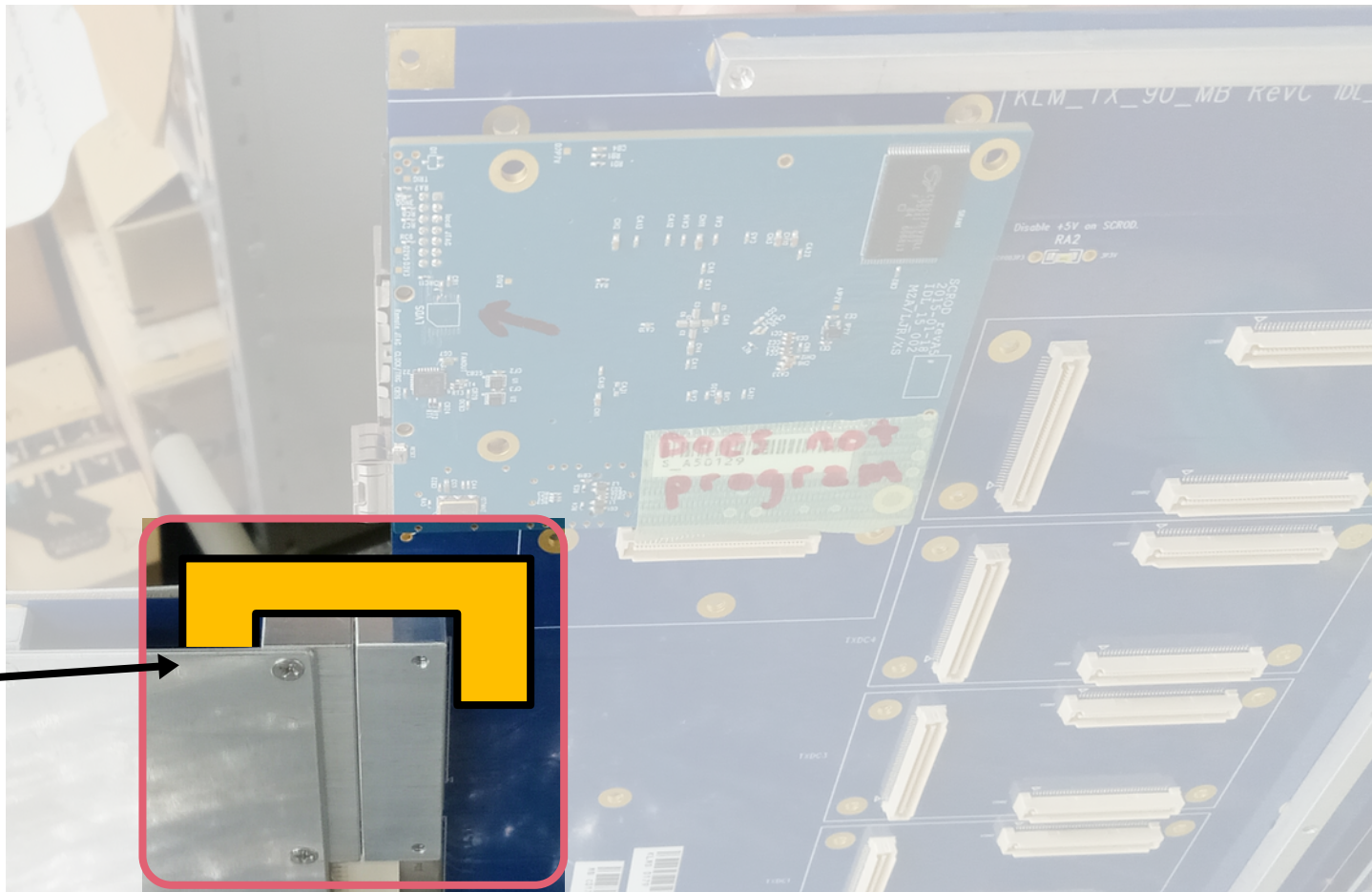
☕ Plan to fabricate clamp to hold motherboard and RHIC together during removal



TABLETOP INVESTIGATION

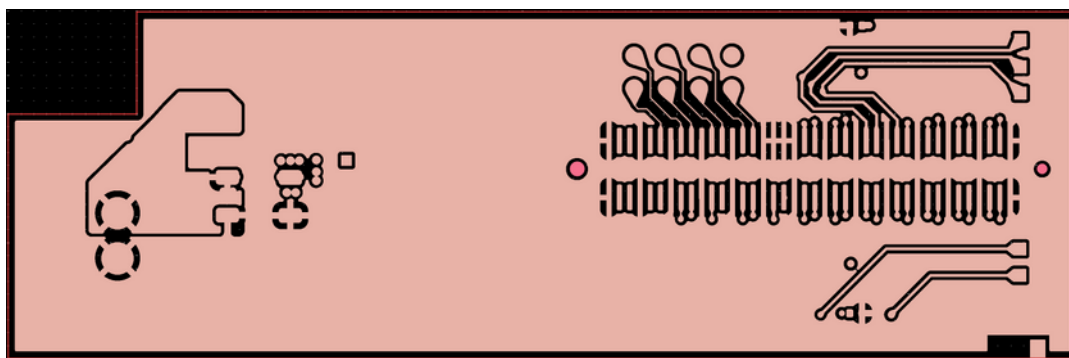
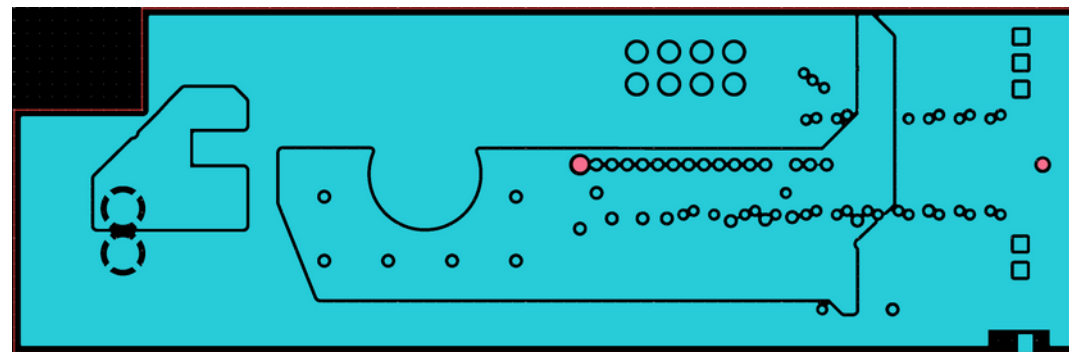
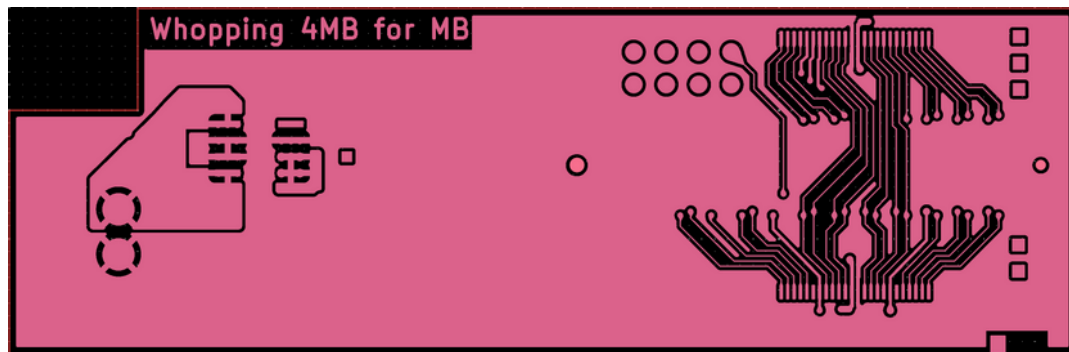
☕ Checked situation using broken FEEs in storage

☕ Plan to fabricate clamp to hold motherboard and RHIC together during removal

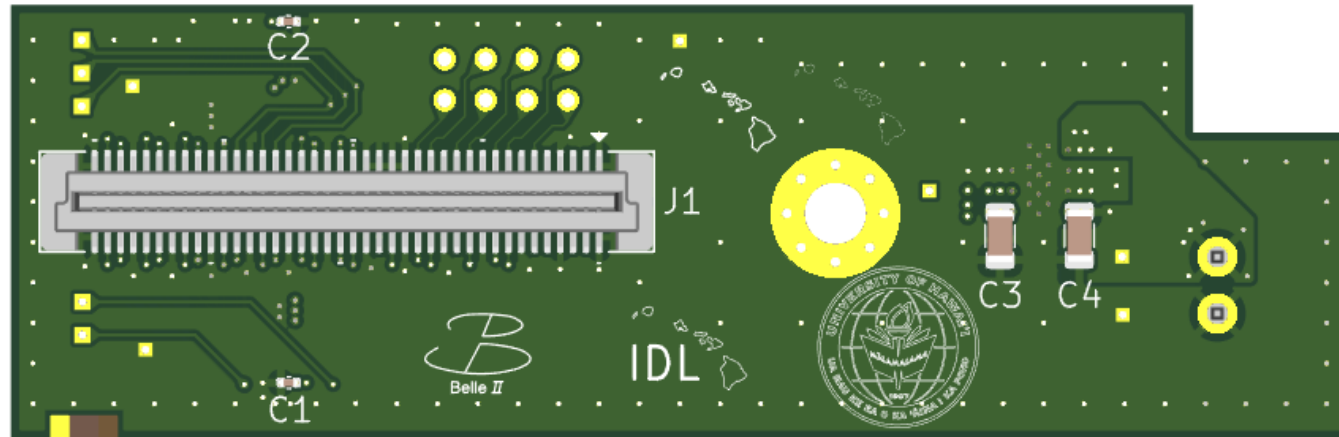
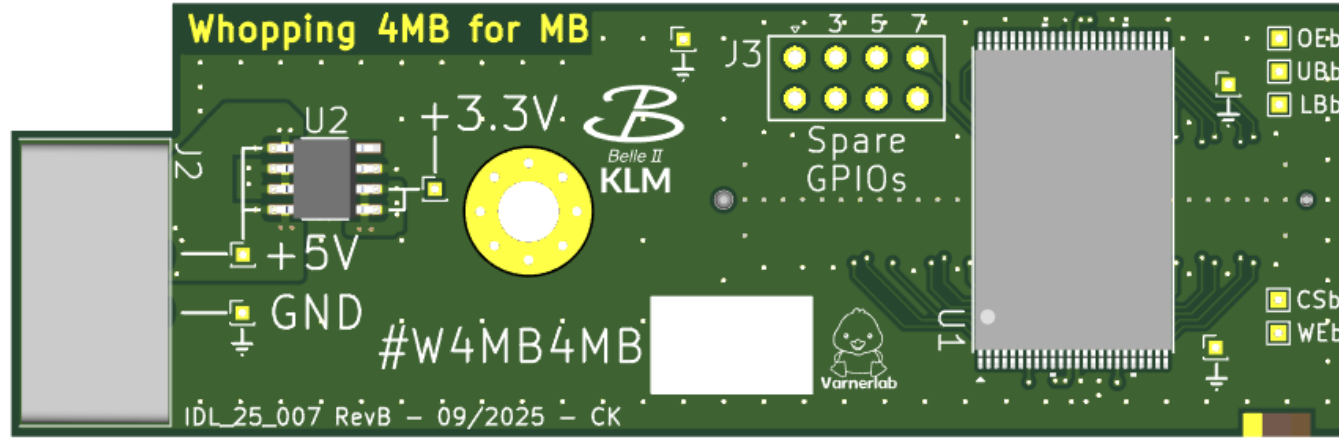


PCB PROGRESS

- ☕ 4-layer PCB
- ☕ No internal routing
- ☕ 1st prototype was delivered today (waiting on SRAM chips)
- ☕ Next:
 - 🔪 Solder prototype and test on Hawaii KLM testbench
 - 🔪 Design rear transition module for +5 V supply

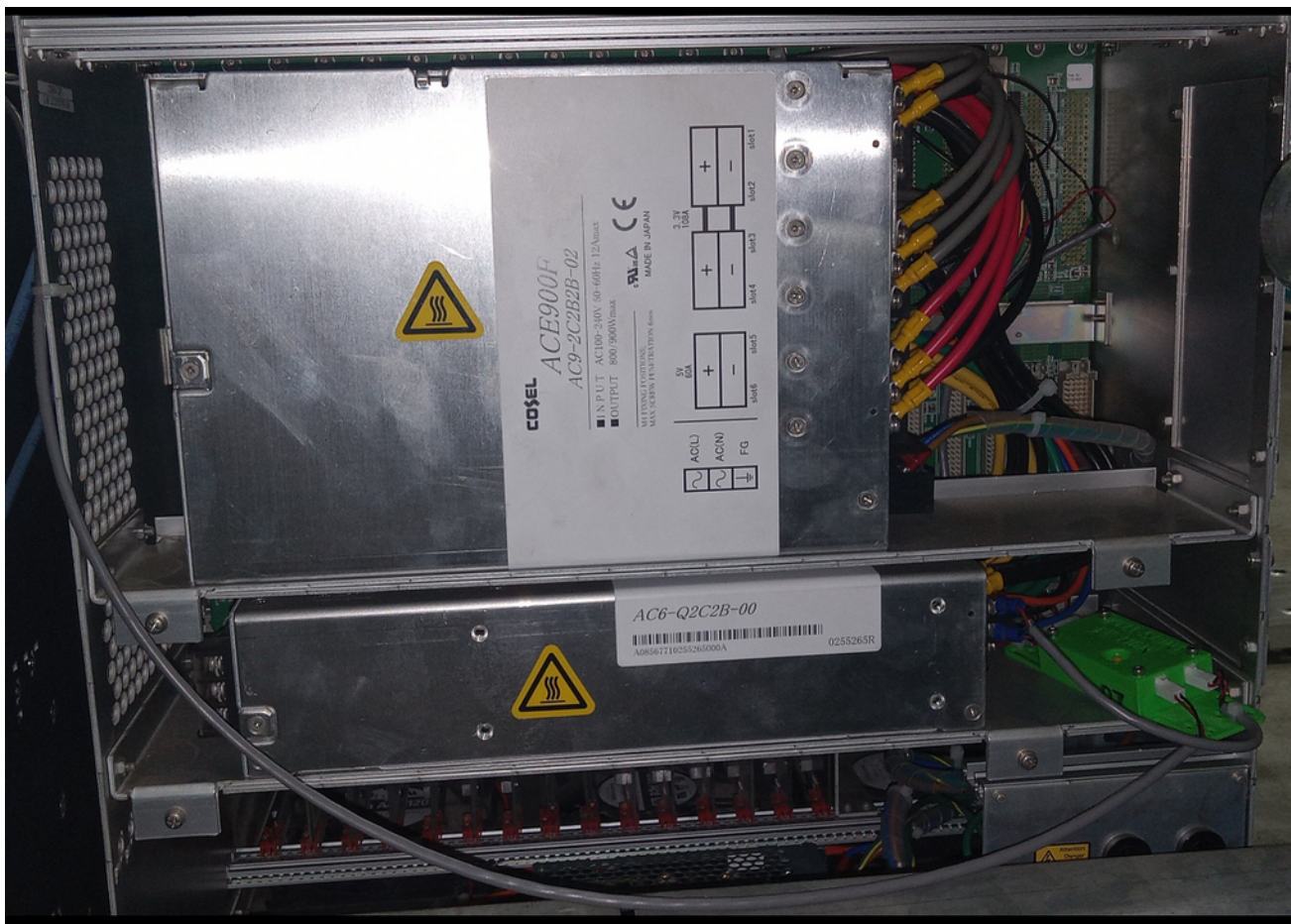


PCB PROGRESS



3.3 V Supply Location

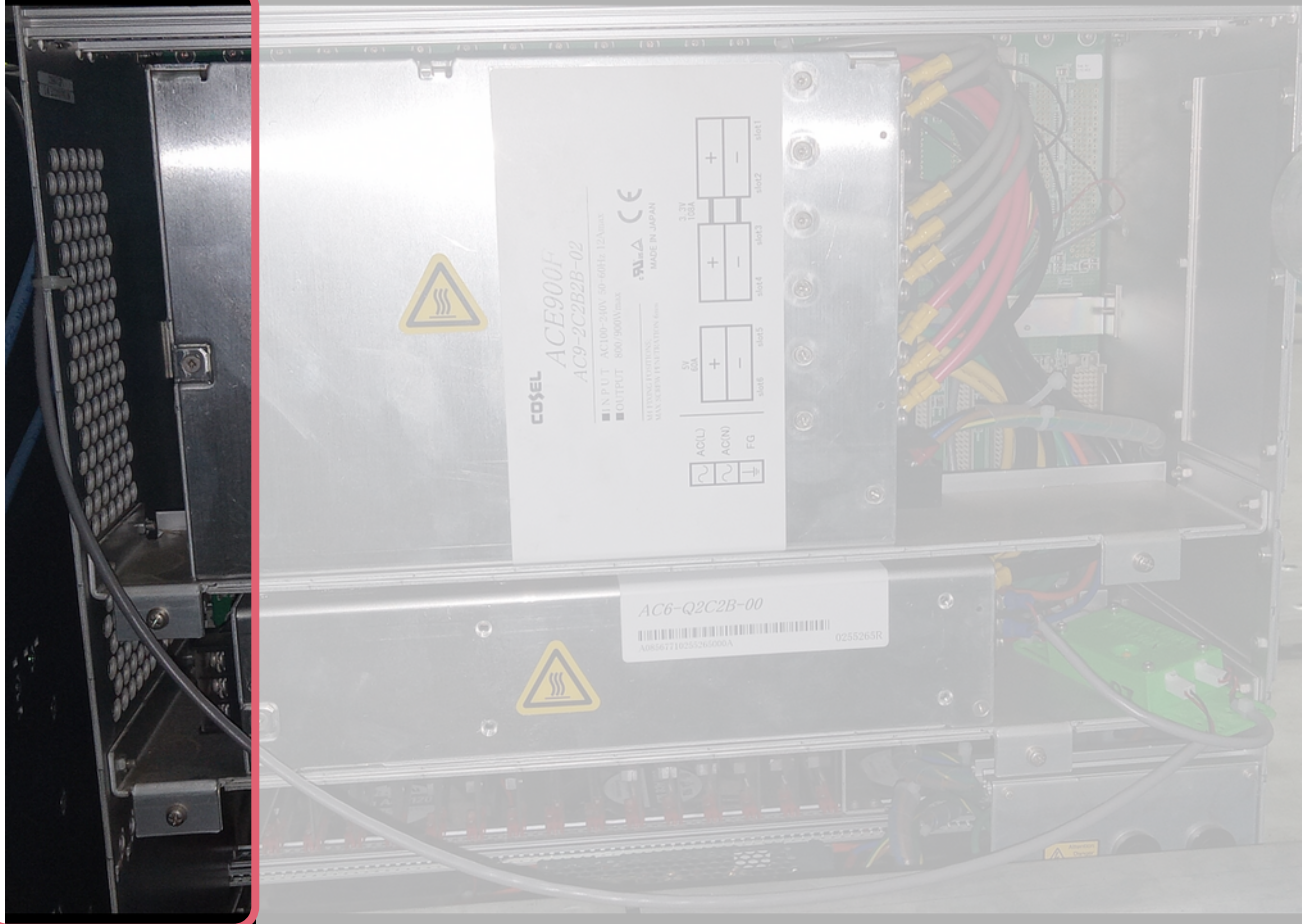
KLM SCINT. CRATE REAR



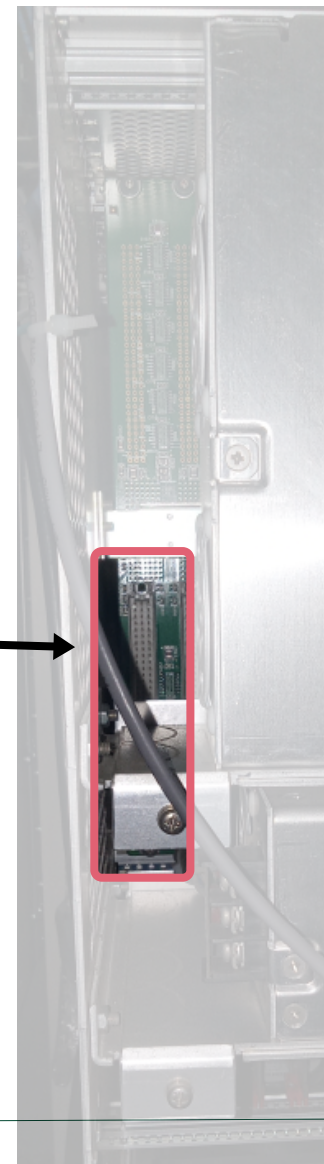
☕ Looking for easy, clean way to get low-voltage power for SRAM expansion boards ...

🕒 Inspected rear of crate for options

KLM SCINT. CRATE REAR

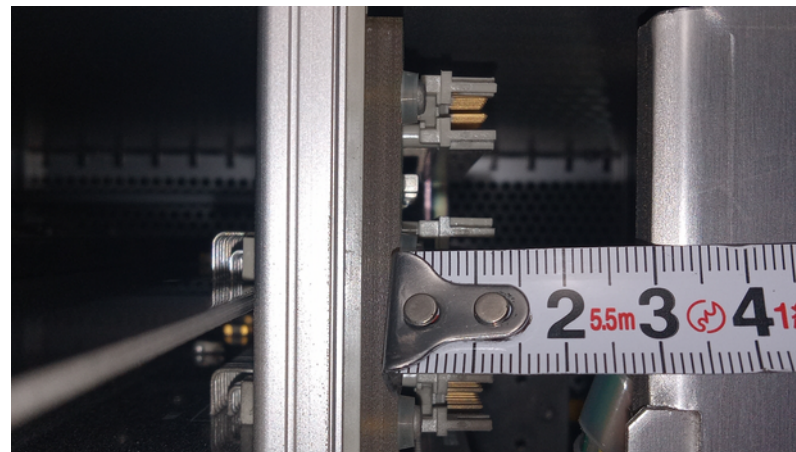


☕ Free connector here with straight shot out of back of crate for a wiring harness



POWER-SUPPLY PLANS

- ☕ Plenty of access space when two Phillips screws are removed from the top power-supply
- ☕ Simple PCB with J1 edge connector
 - 🌰 Max board width 1.8 cm including connector
- ☕ From PCB, use 14-way wire harness with 2-pin connectors at harness positions starting from 112 cm to 140 cm



Firmware

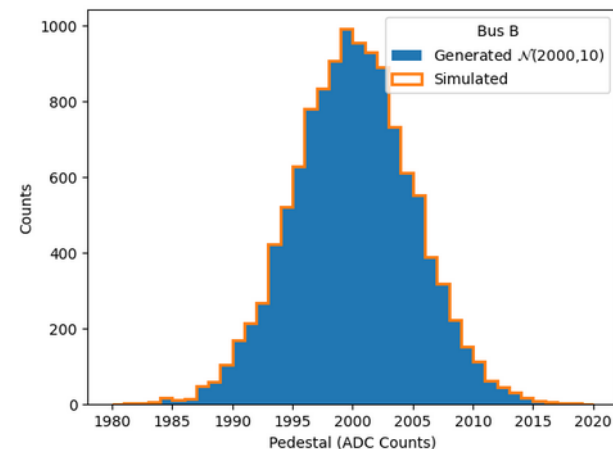
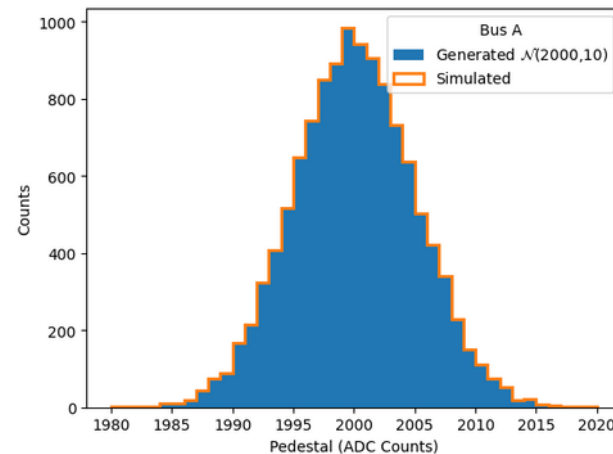
☕ New feature branch **SRAM_upgrade** in
KLM scintillator firmware repository

- ☪ Removed bus A/B arbiter from **PedestalFetcher.vhd** and **PedestalWriter.vhd**
 - Duplicated these "Pedestal" modules and connected one to bus A and one to bus B
- ☪ Connected nets from spare connector on the motherboard to the firmware and cascaded signals down to "Pedestal" modules



Pedestal measurement

- Full (top-level) simulation
- Python script generates random TARGETX ADC data and calculates expected average pedestal measurements for every storage cell
 - Expected 8-bit SRAM values written to text file
- Xilinx ISIM writes measured 8-bit SRAM values to another text file
- Python analysis script compares both text files and plots results



Summary / Next Steps

- ☕ Hardware inspection at KEK validated installation feasibility without uncabling FEEs
- ☕ Acquired measurements for auxiliary components (power distribution)
- ☕ New SCROD firmware feature branch developed and verified in simulation
- ☕ Prototype delivered --- ready for soldering and hardware testing
- ☕ Plan to make 164 SRAM boards and 15 power supply boards (all KLM + 20% spares)
- ☕ Nominal installation planned for next summer

Firmware

WIRE HARNESS DIMENSIONS

☕ 112 to 140 cm (includes 10 cm slack at SCROD side)

- Bend up at 17.5 cm
- Bend fwd after 19.5 cm
- Bend down after 59.5 cm
- First connector after 5.5 + 10 cm
- Additional connector every 2 cm
(7 conn., then 4 cm gap, then 7 more conn.)

☕ Rear power board uses J1/J2 connector

- Board with edge connector should be 1.5 cm wide
- Output harness should egress from the top of the boards rear edge (see right pic. in slide 19)

